Communication via Intents

Mohammed Said BELAID
said.belaid@univ-usto.dz

18/04/2024
@USTO → FMI → Dept Info → L3 SI

Outline

1. Intoduction
2. Types
3. Examples
4. Intent Filters
5. Get a result from an activity

Definition

An Intent is a messaging object that facilitates communication between components within an Android application, as well as between different applications. It serves as a bridge to initiate actions, such as starting an activity, service, or broadcasting a message.

Explicit vs Implicit

- Explicit Intents: Specify a target component within the same application.
- Implicit Intents: Declare the desired action, allowing the system to determine the appropriate component to handle it.

Explicit Intent

- Start a new activity

```
// Explicit Intent to start a new activity
Intent intent = new Intent(context, NewActivity.class);
startActivity(intent);
```

Sending Data with Intent Extras

```
val intent = Intent(this, SecondActivity::class.java).apply {
    putExtra("EXTRA_MESSAGE", "Hello from FirstActivity!")
    putExtra("EXTRA_NUMBER", 42)
}
startActivity(intent)
```

Receiving Data in SecondActivity

```
val message = intent.getStringExtra("EXTRA_MESSAGE")
val number = intent.getIntExtra("EXTRA_NUMBER", 0)
```

Open a Web page

```
val webpage = Uri.parse("https://www.example.com")
val intent = Intent(Intent.ACTION_VIEW, webpage)
startActivity(intent)
```

Share text content

```
val shareIntent = Intent().apply {
    action = Intent.ACTION_SEND
    type = "text/plain"
    putExtra(Intent.EXTRA_TEXT, "Check out this awesome app!")
}
startActivity(Intent.createChooser(shareIntent, "Share via"))
```

Send Email

```
val emailIntent = Intent(Intent.ACTION_SENDTO).apply {
    data = Uri.parse("mailto:") // only email apps should handle
        this
    putExtra(Intent.EXTRA_EMAIL, arrayOf("recipient@example.com")
        )
    putExtra(Intent.EXTRA_SUBJECT, "Subject")
    putExtra(Intent.EXTRA_TEXT, "Message body")
}
startActivity(emailIntent)
```

Dial Phone

```
val phoneNumber = "1234567890"
val dialIntent = Intent(Intent.ACTION_DIAL).apply {
    data = Uri.parse("tel:$phoneNumber")
}
startActivity(dialIntent)
```

Open maps

```
val latitude = 35.69111
val longitude = -0.64167
val geoUri = "geo:$latitude,$longitude"
val intent = Intent(Intent.ACTION_VIEW, Uri.parse(geoUri))
startActivity(intent)
```

Take Photo

```
val takePictureIntent = Intent(MediaStore.ACTION_IMAGE_CAPTURE)
```

Intent Filters

An Intent Filter in Android is a structure used to declare the types of intents that a component (such as an activity, service, or broadcast receiver) can handle. It defines the capabilities of the component and specifies the types of intents it wants to receive.
In the `Manifest`

```
<activity android:name=".MainActivity">
<intent-filter>
  <action android:name="android.intent.action.SEND" />
  <category android:name="android.intent.category.DEFAULT" />
  <data android:mimeType="text/plain" />
</intent-filter>
</activity>
```

Get a result from an activity

You can wait for a response from another activity using:

`startActivityForResult()`

or

`registerForActivityResult()`